35. Scatter Plot Comparing Two Subject Marks Question

Write a Python program to draw a scatter plot comparing two subject marks. **Aim**

To compare the marks of students in two subjects using a scatter plot. **Code**

import matplotlib.pyplot as plt

Marks of students maths = [45, 55, 60, 65,

70, 75, 80, 85, 90, 95] science = [50, 58, 62, 68,

72, 78, 75, 88, 92, 96]

Create scatter plot

plt.scatter(maths, science)

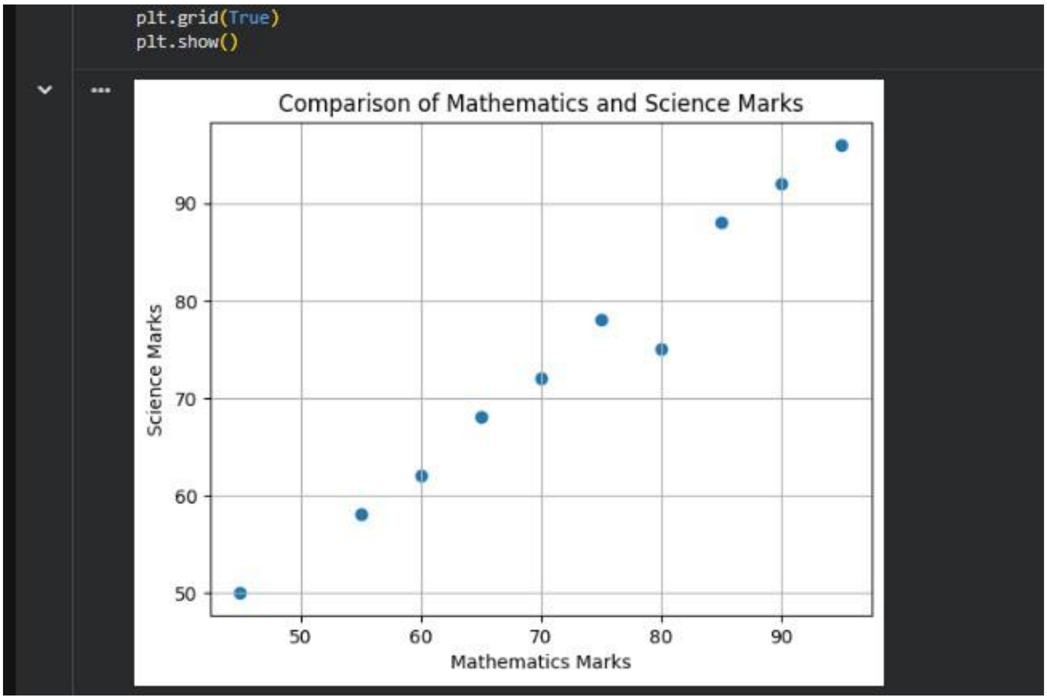
Labels and title plt.xlabel("Mathematics Marks")

plt.ylabel("Science Marks") plt.title("Comparison of

Mathematics and Science Marks")

plt.grid(True)

plt.show() **Output:**



36. Scatter Plot for Three Groups Comparing Weights and Heights

Question

Write a Python program to draw a scatter plot for three different groups comparing weights and heights. **Aim**

To compare the height and weight of people belonging to three different groups using a scatter plot.

Code import matplotlib.pyplot as plt

```
# Group 1 height1 = [150, 155, 160,
165, 170] weight1 = [45, 50, 55, 60,
65]

# Group 2 height2 = [155, 160,
165, 170, 175] weight2 = [50, 55,
60, 65, 70]

# Group 3
height3 = [160, 165, 170, 175, 180]
weight3 = [55, 60, 65, 70, 75]

# Create scatter plots plt.scatter(height1,
weight1, label='Group 1') plt.scatter(height2,
weight2, label='Group 2') plt.scatter(height3,
weight3, label='Group 3')

# Labels and title plt.xlabel("Height (cm)")
plt.ylabel("Weight (kg)") plt.title("Height vs
Weight for Three Groups")

plt.legend()
plt.grid(True)
plt.show()
```

Output:

```
plt.title("Height vs Weight for Three Groups")

plt.legend()
plt.grid(True)
plt.show()
```